

Entry: 1

Item: 5

Question:

Our

first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and hav

Answer:

Producer can produce multiple values. When consumer sleeps for 1 after consumption, the num-full would have temporar

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

13.0

Score 2:

15

Score 3:

13

Entry: 2

Item: 5

Question:

Our first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and

Answer:

By increasing the size of the buffers, the producer has the ability to add multiple values to the buffer (up to its maximum c

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

27.0

Score 2:

27

Score 3:

27

Score outlier:

9

Entry: 3

Item: 5

Question:

What would you predict num_full to be with different buffer sizes?

Answer:

No because there are no other consumers consuming the data asked by other consumers.

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context switches.

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

1.0

Score 2:

0

Score 3:

0

Entry: 4

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

In this setup, the producer produces a value, as soon as mutex unlocks, the consumer consumes the value right away. T

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

2

Score 2:

5

Score 3:

3

Score outlier:

27

Entry: 5

Item: 5

Question:

Our first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and

Answer:

When the buffer is increased, the behavior of the code do not change a lot. Producer produces one then consumer consu

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

2.0

Score 2:

4

Score 3:

3

Entry: 6

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and have the producer produce a few values. Start with a buffer (size 1), and then increase it. How does the behavior of the code change with larger buffers? (or does it?) What would you predict `num_full` to be with different buffer sizes (e.g., -m 10) and different numbers of produced items (e.g., -l 100), when you change the consumer sleep strategy (e.g., -C 0,0,0,0,0,0,1)?

Answer:

If we only change the buffer size the `num_full` will not change, the different number of produced items make the program

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

3

Score 2:

3

Score 3:

3

Entry: 7

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

The enhancement of buffer size does not change the behavior of the code. The `null_full` will first goes from 0 to 1 and the
With `-C 0,0,0,0,0,0,1`, this means the consumer will sleep for some time after consuming each item. This will slow down the

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

4

Score 2:

3

Score 3:

3

Entry: 8

Item: 5

Question:

Our first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and

Answer:

Let -l be constant 3. Increase the buffer size, the program does not change, where only one buffer is used, and the rest is

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

11

Score 2:

15

Score 3:

12

Entry: 9

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

The producer produces a value into buffer with an index of 0 at first. After the consumer consumes it, the producer produ

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

8

Score 2:

16

Score 3:

12

Entry: 10

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

The behavior of the code will become increase in the number of buffers. The prediction of `num_full` should be changed be

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

18

Score 2:

20

Score 3:

21

Entry: 11

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

With larger buffers, the producer is able to add multiple values to the buffer before waiting or switching to another task, and

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

27.0

Score 2:

27

Score 3:

27

Entry: 12

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

Buffer from 1 to 2. The behavior has no change. Buffer from 1 to 5, the behavior has no change too. Because the consumer

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

11

Score 2:

15

Score 3:

12

Entry: 13

Item: 5

Question:

Our

first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and have

Answer:

Behavior Changes :Producers can produce more items before they have to wait for consumers to consume some items ;

Prediction The num_full will increase because the consumers will sleep for 1 second at the c6 , allowing more time for pro

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

15

Score 2:

13

Score 3:

12

Entry: 14

Item: 5

Question:

Our first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and

Answer:

If we increase the larger buffer, it means the producer can add multiple values to the buffer until the buffer is in full, before

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

27.0

Score 2:

25

Score 3:

27

Score outlier:

3

Entry: 15

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

By utilizing larger buffers, the producer gains the ability to add multiple values to the buffer, up to its maximum length, before

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

27.0

Score 2:

27

Score 3:

27

Entry: 16

Item: 5

Question:

Our

first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and have

Answer:

The behavior of the code doesn't change since the default sleeping time for both producer and consumer are zero. No m

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

12

Score 2:

18

Score 3:

15

Entry: 17

Item: 5

Question:

Our first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and

Answer:

When $m=1/2/3$, the result is the same as the increase in buffer size will not affect the production and consumption.

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

7.0

Score 2:

5

Score 3:

6

Entry: 18

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

No, increasing the buffer size does not the behavior of the code in this example. But the larger the buffers, the more values

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

20

Score 2:

16

Score 3:

18

Entry: 19

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

After the producer produces a value, it can point to another empty buffer slot and no need to wait for the value to be consumed.

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context switches.

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

9

Score 2:

11

Score 3:

9

Score outlier:

27

Entry: 20

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

When using larger buffers, the code exhibits a change in behavior as the producer can store more values in the buffer before

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

27.0

Score 2:

27

Score 3:

27

Entry: 21

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

There is no significant difference in code change to larger buffers. The user will consume the same values from the buffer

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

15.0

Score 2:

10

Score 3:

12

Entry: 22

Item: 5

Question:

Our

first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and hav

Answer:

With a larger buffer producer will be able to produce more (up to buffer capacity) before waiting for the consumer or a con

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

23.0

Score 2:

16

Score 3:

18

Entry: 23

Item: 5

Question:

Our first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and

Answer:

num_full = 3 when -m 10 is used num_full = 100 when -l 100 is used They seem to don't change even when there are sleep

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

6

Score 2:

6

Score 3:

3

Entry: 24

Item: 5

Question:

Our

first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and hav

Answer:

The producer can add many different values to the buffer (the value is belong to $[1, \text{size of buffer} - 1]$, depending on at wh

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

20

Score 2:

27

Score 3:

21

Entry: 25

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

The result does not change when the buffers become larger. When `num_full` changes or consumer sleep string changes,

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

5

Score 2:

9

Score 3:

3

Entry: 26

Item: 5

Question:

Our

first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and have

Answer:

With larger buffers, the producer can store multiple values in the buffer before pausing or switching tasks, and the same a

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

27

Score 2:

27

Score 3:

27

Score outlier:

6

Entry: 27

Item: 5

Question:

Our

first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and have

Answer:

num_full changes only if the consumer sleeps at least once. Otherwise, it's either 0 or 1.

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

0.0

Score 2:

1

Score 3:

0

Entry: 28

Item: 5

Question:

Our first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and

Answer:

Changes■ The producer can add multiple values to the buffer (up to the length of the buffer -1) before it waits, and vice v

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

27.0

Score 2:

27

Score 3:

27

Entry: 29

Item: 5

Question:

Our

first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and have

Answer:

With a buffer size (`-m 10`) and a larger number of produced items (`-l 100`), the `num_full` will be `num_full`; would fluctuate based on the relative speeds of production and consumption. After consumer sleeps

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

12.0

Score 2:

12

Score 3:

12

Entry: 30

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

`Num_full` is always either be 9 or 10 instead of 1. Because consumer have to wait for 1 second and within this second, pr

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

9.0

Score 2:

10

Score 3:

9

Entry: 31

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

With a larger buffer (from $m=1$ to $m=3$), the behavior remains unchanged. For no sleep, the value of `num_full` only changes

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

10

Score 2:

12

Score 3:

12

Entry: 32

Item: 5

Question:

Our

tag{first question} focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and

Answer:

The behavior of the code will not change the with the larger buffer. num_full always is 0 and 1. When the consumer sleeps

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

4

Score 2:

3

Score 3:

0

Entry: 33

Item: 5

Question:

Our

first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and hav

Answer:

The behavior of the code does not change with larger buffers as producer and consumer take turns to acquire and releas

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

13

Score 2:

15

Score 3:

13

Entry: 34

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

The buffer size is affecting the synchronization between different threads. With a bigger buffer size, more data can be pro

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

16

Score 2:

10

Score 3:

12

Entry: 35

Item: 5

Question:

Our

first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and hav

Answer:

When the buffer is larger or more numbers of produced items, the num_full is always 0 or 1, which the number of entries

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or contex

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

5

Score 2:

5

Score 3:

2

Entry: 36

Item: 5

Question:

Our

first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and have

Answer:

A larger buffer can also provide more space to let the producer produce more value before the consumer consumes, same as the consumer can take turns.

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context switches.

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

27.0

Score 2:

27

Score 3:

27

Entry: 37

Item: 5

Question:

Our first question focuses on `main-two-cvs-while.c` (the working solution). Run with one producer and one consumer, and

Answer:

The behavior of the code changes with larger buffers as the producer is able to produce more items before the consumer

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

9.0

Score 2:

12

Score 3:

9

Entry: 38

Item: 5

Question:

Our first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and

Answer:

If the buffer is size 1, the user_ prt and the fill_ prt always point at the same address. The larger buffer code has different

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

2

Score 2:

0

Score 3:

0

Entry: 39

Item: 5

Question:

Our first question focuses on main-two-cvs-while.c (the working solution). Run with one producer and one consumer, and

Answer:

A larger buffer allow the producer to produce more without waiting the consumer to consume ,as long as there is still space

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

9.0

Score 2:

8

Score 3:

9

Entry: 40

Item: 5

Question:

Our

have the consumer consume a value, have the producer produce a value Okay, so now we have

Here's But

on p0, on c0 , on p1, on c0, new value of p1, old value of p0; now on p4, on c1, no space here, on p5, on c4, no space h

[new value of p6, example value of p1, old value of p4, old value of p5, old value of p1, on c0, now on c1] 0,

on c5, no the c0,

on c5

, on c4,

on c1

executable

producer -c 0,0,0,0,0,0,1

Answer:

./main-two-cvs-while -p 1 -c 1 -m 1 -v -t NF P0 C0 0 [*---] p0 0 [*---] c0 0 [*---] p1 1 [* 0] p4 1 [* 0] p5 1 [* 0] p6 1 [* 0] c

Sample answer:

With larger buffers, producer can add multiple values to the buffer (up to the length of the buffer) before it waits or context

Sample criteria:

3 sub-questions, 9 points for each sub-question

Full points:

27

Score 1:

5

Score 2:

4

Score 3:

3